Line counts on fixed sets of boxes: a polynomial tail, and the strip-cover scheme

Working note

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Abstract

For a Poisson line process of intensity μ and an arbitrary fixed set W of m unit boxes, we prove that the total number of line–box incidences on W exceeds κm with probability at most $Cm^{-c\kappa}$, for all $\kappa \geq \kappa_0(\mu)$. The proof decomposes the lines by the size of their footprint on W, bounds the mass of long footprints by a pair count, and applies Poisson tail estimates class by class. For a straight row of boxes the lines with long footprints are automatically nearly parallel to the row, which is the structural statement needed for the thick-cluster lemma of the companion note. We then describe a Peierls argument over covers of a thick cluster by straight strips, which applies the fixed-set bound to the strips and thereby avoids the entropy of paths inside bundles, and we isolate the one estimate it still requires: a product bound for the incidence events of nearly parallel strips. The single-cluster bound, and hence the unconditional subcritical theorem, remain open.

1 Setting

Let $\mathcal{L}$ be a Poisson line process with intensity measure $\nu = \nu_\mu$, $\nu(\mathrm{d}\theta\,\mathrm{d}s) = (\mu/\pi)\,\mathrm{d}\theta\,\mathrm{d}s$. For $z \in \mathbb{Z}^2$ let $Q(z) = z + [0,1)^2$ and let $N(z)$ be the number of lines meeting $Q(z)$, a Poisson variable with mean $\mu_1 := 4\mu/\pi$. For a finite set $W \subseteq \mathbb{Z}^2$ of m boxes put

$$S_W := \sum_{z\in W} N(z) = \sum_{\ell\in\mathcal{L}} f_W(\ell), \qquad f_W(\ell) := \#\{z \in W : \ell \cap Q(z) \neq \varnothing\},$$

the number of line–box incidences on W; $f_W(\ell)$ is the *footprint* of ℓ on W. We recall from [HJ26c, Lemma 3.2] that two unit boxes whose centres are at Euclidean distance D are both met by lines of ν-measure at most $2\mu/D$ when $D \geq 2\sqrt{2}$, and at most μ_1 always; hence

$$\nu(\{\ell : \ell \text{ meets } Q(z) \text{ and } Q(z')\}) \leq \frac{6\mu}{1+|z-z'|} \qquad (z \neq z'). \tag{1.1}$$

The companion note [HJ26a] reduces the subcritical phase, for $\lambda < 1/(576\sqrt{2}(j_0-1))$, to a thick-cluster lemma whose core is the single-cluster bound: for a 2-connected set A of d boxes

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through a fixed box, all boxes of which are met by at least j_0 lines,

$$\mathbb{P}\big(\exists\, 2\text{-connected } A \ni 0,\ |A| = d,\ \min_{z\in A} N(z) \ge j_0\big) \le C\, d^{-\beta}, \qquad \beta > 4\,. \tag{1.2}$$

The present note proves the analogous bound for a *fixed* set W in place of the union over A, which is the first of the two steps proposed there, and sets up the second.

2 The fixed-set tail

Theorem 2.1 (Polynomial tail for fixed sets)**.** *There are constants $\kappa_0 = \kappa_0(\mu)$, $C = C(\mu,\kappa)$ and a universal $c > 0$ such that for every finite set $W \subseteq \mathbb{Z}^2$ of $m \ge 2$ boxes and every $\kappa \ge \kappa_0$,*

$$\mathbb{P}(S_W \ge \kappa m) \le C\, m^{-c\kappa}\,. \tag{2.1}$$

One may take $c = 1/80$ and $\kappa_0 = 5 \cdot 10^4 \mu$.

The constants are not optimised. The bound is polynomial because of lines whose footprint is comparable to m: for a straight row W, κ nearly parallel lines along the row produce $S_W \ge \kappa m$ with probability of order $m^{-(\kappa-1)}$, so an exponent linear in κ is the correct order. For sets that are not row-like the true tail is much smaller, and the theorem is correspondingly lossy there.

2.1 Footprint classes

Fix W with $|W| = m$ and write $f = f_W$. Let K be the integer with $2^K \le m < 2^{K+1}$, and for $0 \le k \le K$ let $\mathcal{C}_k := \{\ell\colon 2^k \le f(\ell) < 2^{k+1}\}$, the *class-$k$ lines*; every line with $f(\ell) \ge 1$ belongs to exactly one class. Let $n_k := \#(\mathcal{L} \cap \mathcal{C}_k)$ and $\nu_k := \nu(\mathcal{C}_k)$, so that n_k is Poisson with mean ν_k and

$$S_W \le \sum_{k=0}^{K} n_k 2^{k+1}\,. \tag{2.2}$$

Lemma 2.2 (Mass of the classes)**.** *For every W with $|W| = m$,*

$$\sum_{k=0}^{K} \nu_k 2^k \le \int f\,\mathrm{d}\nu = \mu_1 m, \qquad \nu_k \le \nu(f \ge 2^k) \le 144\,\mu\, m^{3/2}\, 4^{-k} \quad (k \ge 1)\,. \tag{2.3}$$

Proof. The first inequality is $\int f\,\mathrm{d}\nu = \sum_{z\in W} \nu(\ell \text{ meets } Q(z)) = \mu_1 m$. For the second, $\int f(f-1)\,\mathrm{d}\nu = \sum_{z\neq z'\in W} \nu(\ell \text{ meets } Q(z) \text{ and } Q(z')) \le \sum_{z\in W}\sum_{z'\in W\setminus\{z\}} 6\mu/(1+|z-z'|)$ by (1.1). For fixed z, the number of lattice points at distance in $[j-1, j)$ from z is at most $8j$ for $j \ge 1$, so the inner sum is at most $\sum_{j\le R} 8j/j = 8R$, where R is the smallest integer with $\sum_{j\le R} 8j \ge m$, i.e. $R \le \sqrt{m/4} + 1 \le 1.5\sqrt{m}$. Hence $\int f(f-1)\,\mathrm{d}\nu \le 72\mu m^{3/2}$, and for $s \ge 2$, Markov's inequality gives $\nu(f \ge s) \le \int f(f-1)\,\mathrm{d}\nu/(s(s-1)) \le 144\mu m^{3/2}/s^2$. □

The second bound in (2.3) is sharp up to constants for a disk, where every line has footprint of order $\sqrt{m}$, and loses a factor $\sqrt{m}$ for a straight row, where $\nu(f \ge s) \asymp \mu m/s^2$ (Section 2.3). It suffices for a polynomial tail in both cases.

2.2 Thresholds and Poisson tails

Proof of Theorem 2.1. Put $c_1 := 1/20$ and, for $0 \le k \le K$,

$$a_k := \max\{2\mathrm{e}\nu_k,\ c_1\kappa 2^{(K-k)/2}\}, \qquad \theta_k := a_k 2^{k+1}. \tag{2.4}$$

By (2.3) and $2^{K+1} \le 2m$,

$$\sum_{k=0}^{K} \theta_k \le 4\mathrm{e}\sum_k \nu_k 2^k + c_1\kappa 2^{K/2+1}\sum_{k=0}^{K} 2^{k/2} \le 4\mathrm{e}\mu_1 m + \frac{c_1\kappa\, 2^{K+3/2}}{1-2^{-1/2}} \le 4\mathrm{e}\mu_1 m + 9.7\, c_1\kappa m \le \frac{3}{4}\kappa m$$

provided $\kappa \ge 16\mathrm{e}\mu_1$. Hence, on $\{S_W \ge \kappa m\}$, by (2.2) some class satisfies $n_k 2^{k+1} > \theta_k$, that is $n_k > a_k$, and

$$\mathbb{P}(S_W \ge \kappa m) \le \sum_{k=0}^{K} \mathbb{P}(n_k > a_k) \le \sum_{k=0}^{K} \Big(\frac{\mathrm{e}\nu_k}{a_k}\Big)^{a_k}, \tag{2.5}$$

using $\mathbb{P}(\mathrm{Poi}(\nu) \ge a) \le (\mathrm{e}\nu/a)^a$ for $a \ge \nu$, which applies since $a_k \ge 2\mathrm{e}\nu_k$. Because $a_k \ge 2\mathrm{e}\nu_k$, every term is at most $2^{-a_k} \le 2^{-c_1\kappa 2^{(K-k)/2}}$.

Let $k_1 := \lceil 2K/3 \rceil$. For $k < k_1$ we have $2^{(K-k)/2} \ge 2^{K/6} \ge (m/2)^{1/6}$, so these at most $K+1 \le 2\log_2 m$ terms contribute at most $2\log_2 m \cdot 2^{-c_1\kappa(m/2)^{1/6}}$, which is $O(m^{-\kappa})$.

For $k \ge k_1$ we use the second bound in (2.3) together with $a_k \ge c_1\kappa 2^{(K-k)/2}$ and $m^{3/2} \le 2^{3(K+1)/2}$:

$$\frac{\mathrm{e}\nu_k}{a_k} \le \frac{144\mathrm{e}\mu\, 2^{3(K+1)/2} 4^{-k}}{c_1\kappa 2^{(K-k)/2}} = \frac{144\mathrm{e}2^{3/2}\mu}{c_1\kappa}\, 2^{K-3k/2} =: r_k .$$

Writing $k = K - j$ with $0 \le j \le K - k_1 \le K/3 + 1$, we get $r_{K-j} = \gamma\, 2^{-K/2} 2^{3j/2}$ with $\gamma := 144\mathrm{e}2^{3/2}\mu/(c_1\kappa) = 22144\mu/\kappa$, and $a_{K-j} \ge c_1\kappa 2^{j/2}$. For $\kappa \ge \kappa_0 = 5\cdot 10^4\mu$ we have $\gamma \le 1/2$, and since $j \le K/3 + 1$ we have $2^{3j/2} \le 2^{K/2+3/2}$, so $r_{K-j} \le 2^{3/2}\gamma \le 2^{1/2}$; the terms with $r_{K-j} \ge 1/2$ are still bounded by 2^{-a_k} from the first remark, which gives $2^{-c_1\kappa 2^{j/2}}$ for those j, and these j satisfy $2^{3j/2} \ge 2^{K/2}/(2\gamma)$, i.e. $2^{j/2} \ge (2^{K/2}/(2\gamma))^{1/3} \ge (m/2)^{1/6}$, so their total contribution is again $O(m^{-\kappa})$. For the remaining j, $r_{K-j} < 1/2$, and

$$\Big(\frac{\mathrm{e}\nu_k}{a_k}\Big)^{a_k} \le r_{K-j}^{\,c_1\kappa 2^{j/2}} = \big(\gamma 2^{-K/2}\big)^{c_1\kappa 2^{j/2}} 2^{(3j/2)c_1\kappa 2^{j/2}}.$$

With $x := \gamma 2^{-K/2} \le \gamma\sqrt{2/m}$, the j-th term is at most $\exp\{c_1\kappa 2^{j/2} h(j)\}$ with $h(j) := \log x + 1.04 j$, where $1.04 \ge \frac{3}{2}\log 2$. On the range where $r_{K-j} < 1/2$ we have $2^{3j/2} < 2^{K/2}/(2\gamma)$, hence $1.04 j \le \frac{1}{2}\log m + \log(1/(2\gamma))$ and $h(j) \le \log(\gamma\sqrt{2}) + \log(1/(2\gamma)) = -\frac{1}{2}\log 2 < -0.34$. Put $j^* := (-\log x)/2.08$. For $j \le j^*$ we have $h(j) \le \frac{1}{2}\log x$, so the term is at most $x^{c_1\kappa 2^{j/2}/2}$, and these terms are bounded by a geometric series with sum at most $2x^{c_1\kappa/2}$. For $j > j^*$ we have $2^{j/2} \ge 2^{j^*/2} \ge (1/x)^{1/6} \ge (m/2)^{1/12}\gamma^{-1/6}$, and $h(j) < -0.34$, so each of these at most K terms is at most $\exp\{-0.34 c_1\kappa (m/2)^{1/12}\}$, which is $O(m^{-\kappa})$ in total.

Collecting the contributions, $\mathbb{P}(S_W \ge \kappa m) \le 2x^{c_1\kappa/2} + O(m^{-\kappa}) \le 2(\gamma\sqrt{2})^{c_1\kappa/2} m^{-c_1\kappa/4} + O(m^{-\kappa})$, which gives (2.1) with $c = c_1/4 = 1/80$, and a fortiori with $c = 1/140$. □

Remark 2.3. The proof uses only two properties of the footprint measure: $\int f\,\mathrm{d}\nu \le \mu_1 m$, and $\nu(f \ge s) \le C\mu m^{3/2}/s^2$. Any random environment whose line–box incidences satisfy these two bounds, with the line counts of disjoint sets of lines independent, obeys the same tail. The pair bound (1.1) is the only geometric input.

2.3 Straight rows and nearly parallel lines

For a straight row $R = \{(i,0)\colon 0 \le i < m\}$ the footprints have a simpler structure, which gives the "nearly parallel groups" statement of step (i) in [HJ26a, Section 5].

Lemma 2.4 (Rows)**.** *Let R be a straight row of m boxes and let ℓ make the angle $\alpha \in [0,\pi/2]$ with the row direction. Then $f_R(\ell) \le \min\{m,\ 1/\sin\alpha + 2\}$. Consequently every line with $f_R(\ell) \ge s \ge 3$ satisfies $\sin\alpha \le 1/(s-2)$, and $\nu(f_R \ge s) \le 3\mu m/(s-2)^2 + \mu/(s-2)$ for $s \ge 3$.*

Proof. The line meets the strip $[0,m]\times[0,1]$ in a segment whose horizontal extent is at most $1/\tan\alpha \le 1/\sin\alpha$, so it meets at most $1/\sin\alpha + 2$ boxes of the row. Lines with direction angle in $[\alpha,\alpha+\mathrm{d}\alpha]$ meeting the strip have ν-measure $(\mu/\pi)(m\sin\alpha+\cos\alpha)\,\mathrm{d}\alpha$, the width of the strip in the normal direction; integrating over $\sin\alpha \le 1/(s-2)$ gives at most $(\mu/\pi)(m\alpha_s^2/2\cdot(1+o(1))+\alpha_s)\cdot 2$ with $\alpha_s \le \pi/(2(s-2))$, hence the stated bound. □

Thus for a row the lines of class k are exactly those within angle $O(2^{-k})$ of the row: long footprints come from nearly parallel lines, and Theorem 2.1 for a row is a statement about bundles of nearly parallel lines. With the sharper measure bound of Lemma 2.4 in place of (2.3), the proof of Theorem 2.1 gives the exponent $c\kappa$ with $c = 1/40$ and $\kappa_0 = 2\cdot 10^3\mu$ for rows; we omit the arithmetic.

3 The strip-cover scheme

Theorem 2.1 bounds the tail for one set. The single-cluster bound (1.2) needs a union over all 2-connected sets of size d through the origin, of which there are up to 24^d, and a union bound with (2.1) fails: inside a bundle of j_0 lines of length d, of probability $\asymp (\mu/d)^{j_0-1}$, every one of the $\ge 2^d$ paths in the bundle's strip is thick. The entropy must be organised around low-entropy sets to which Theorem 2.1 applies. We describe the scheme and the estimate it needs.

Strips. A *strip* is a set $T = T(x,u,t) \subseteq \mathbb{Z}^2$ of the boxes within sup-norm distance 2 of the segment $\{x+su\colon 0\le s\le t\}$, $x\in\mathbb{Z}^2$, u a unit vector, $t \ge 1$; it has at most $5t+25$ boxes. Up to the discretisation of u to precision $1/t$, which changes T by $O(1)$ boxes per unit length and is absorbed in the width, the number of strips of length t with a given starting box is at most Ct.

Covers. Let $A \ni 0$ be 2-connected with $|A| = d$. Walk along a spanning 2-path of A with at most $2d$ steps and cut it into maximal pieces that stay inside a strip of their own direction: a piece is a sub-path contained in $T(x,u,t)$ with t comparable to the piece's displacement. Each piece has length at most $2t + C$, and the pieces' strips $T_1,\dots,T_s$ cover A with $\sum_i |T_i| \le C\sum_i t_i \le Cd$; consecutive strips start within distance 2 of the end of the previous one. Short pieces, of length at most t_0, are kept as they are. The data $(u_i,t_i)_{i\le s}$ determine the cover, with entropy at most $\prod_i (Ct_i)$ for long pieces and 24^{t_0} for short ones. The point of the construction is that the walk is free to wiggle inside each strip: its wiggles are not part of the data.

Cost of a strip. If every box of A is thick, then $S_{T_i} \ge \sum_{z\in A\cap T_i} N(z) \ge j_0|A\cap T_i| \ge j_0 t_i$, because the piece inside T_i visits at least t_i distinct boxes. By Theorem 2.1 applied to the fixed set T_i, with $m = |T_i| \le 5t_i + 25$ and $\kappa = j_0 t_i/m \ge j_0/30$,

$$\mathbb{P}(S_{T_i} \ge j_0 t_i) \le C\, t_i^{-cj_0/30} \qquad (t_i \ge t_0)\,, \tag{3.1}$$

and for short pieces, every box thick costs at least the probability $\varepsilon_0 := (\mathrm{e}\mu_1/j_0)^{j_0}$ that one given box is thick.

Peierls sum, if the strips were independent. With $w_t := Ct \cdot Ct^{-cj_0/30}$ for $t \geq t_0$ and $w_t := 24^t\varepsilon_0$ for $t < t_0$, the sum over covers of the product of the strip costs is bounded by the coefficient of x^d in $\sum_{s\geq 1} g(x)^s$ with $g(x) = \sum_t w_t x^t$. If $cj_0/30 > 2$ and j_0 is large enough that $g(1) < 1/2$, then this coefficient is at most $Cd^{1-cj_0/30}$: a cover reaching total length d with s pieces has a piece of length at least d/s, and $\sum_s s\, w_{d/s}\, g(1)^{s-1} = O(d^{1-cj_0/30})$. This is the single-cluster bound (1.2) with $\beta = cj_0/30 - 1$, which exceeds 4 for $j_0 \geq 150/c$.

What is missing. The strip events $\{S_{T_i} \geq j_0 t_i\}$ are not independent: a line may have a long footprint on several strips. The product bound $\mathbb{P}(\bigcap_i \{S_{T_i} \geq j_0 t_i\}) \leq \prod_i \mathbb{P}(S_{T_i} \geq j_0 t_i)$ is therefore not justified as it stands. Two observations limit the dependence.

Lemma 3.1 (Long footprints select directions)**.** *Let $T = T(x, u, t)$ and $T' = T(x', u', t')$ be strips whose directions make an angle $\alpha \geq 10/F$, and let ℓ be a line. Then $\min\{f_T(\ell), f_{T'}(\ell)\} \leq F$.*

Proof. A line meeting more than F boxes of T has a segment of length at least $F/5 - C$ inside a strip of width 5, hence makes an angle at most $\arcsin(5/(F/5 - C)) \leq 30/F$ with u for $F \geq C'$; the same for T'. Both cannot hold if the directions differ by more than $60/F$; the constant 10 in the statement is adjusted accordingly. □

In the proof of Theorem 2.1 only the classes $k \geq k_1$, i.e. footprints at least $m^{2/3}$, contribute the polynomial part of the tail; the lower classes are controlled by Poisson tails that are stretched-exponential in m and depend only on the total measure of short-footprint lines, which is dominated by the number of lines meeting the strip and is handled uniformly as in [HJ26a, Lemma 3.2]. Hence the dependence that matters is through lines with footprint at least $t_i^{2/3}$ on strip i, and by Lemma 3.1 such a line can serve two strips only if their directions are within angle $O(t^{-2/3})$. For strips in distinct direction classes the long-footprint line sets are disjoint, and van den Berg's disjoint-occurrence inequality for Poisson processes [vdB96], in the form used in the companion corridor notes [HJ26e], gives the product bound for the long-footprint parts of the events. What remains is the following.

Definition 3.2 (Parallel-strip hypothesis)**.** Hypothesis (P) holds if for every family of strips $T_1, \ldots, T_s$ with pairwise nearly parallel directions (within angle $t^{-2/3}$ of a common u, $t = \min t_i$) and starting boxes along a common walk as above, the probability that all of them have $S_{T_i} \geq j_0 t_i$ is at most $C^s \prod_i \mathbb{P}(S_{T_i} \geq j_0 t_i)^{1/2}$.

The exponent $1/2$ is a placeholder for any uniform positive power; it halves the exponent in (3.1) and is harmless. Hypothesis (P) is the strip-level version of the "joined corridors" estimates of [HJ26b, HJ26e], which prove exactly such product bounds for corridors sharing one supporting line, and of the lower bound of [HJ26d], which shows that a factor C^s cannot be avoided for oscillating families. In the present setting the oscillating families are not dangerous, since their strips have average line count of order 2 rather than j_0, but a proof of (P) must still exclude them explicitly.

Proposition 3.3 (Conditional single-cluster bound)**.** *If (P) holds, then (1.2) holds with $\beta = cj_0/60 - 1$ for all $j_0 \geq 300/c$, and consequently, by [HJ26a, Theorem 1.2] and the chain bookkeeping of [HJ26a, Section 3.1], $\lambda_c(\mu, 1) > 0$ for all $\mu \leq \mu_0$.*

Proof sketch. Group the strips of a cover into direction classes of width $t^{-2/3}$; for distinct classes use the disjoint-occurrence inequality on the long-footprint parts and the uniform short-footprint bound; inside a class use (P). The Peierls sum then runs as above with w_t replaced by $C\,t \cdot t^{-cj_0/60}$, and the chain bookkeeping of [HJ26a] converts the single-cluster bound into hypothesis (T) there. □

4 Summary

1. Theorem 2.1 is rigorous: for any fixed set of m boxes the incidence count exceeds κm with probability $O(m^{-c\kappa})$, $c = 1/80$, for $\kappa \geq 5 \cdot 10^4 \mu$. This is step (ii) of the plan in [HJ26a, Section 5], for arbitrary sets rather than straight bundles only.
2. Lemma 2.4 is step (i) for rows: long footprints on a row come from lines within angle $O(1/s)$ of the row, with measure $O(\mu m/s^2)$.
3. The strip-cover scheme of Section 3 replaces the union over connected sets by a union over covers by straight strips and applies Theorem 2.1 to the strips. It removes the entropy of paths inside bundles. Its one missing input is the parallel-strip hypothesis (P), a product bound for nearly parallel strips.
4. Under (P), Proposition 3.3 gives the single-cluster bound with $\beta = cj_0/60 - 1$ and, through [HJ26a], a subcritical phase in the sparse regime. We estimate (P) to be of the same nature as the joined-corridor estimates of the companion notes and therefore within reach, but it is not proved here.

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Use of AI tools

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